

Agent: host_agent

The main tidal water features near Exeter are:

- Exeter Canal (Named Canal, length: 8,281 meters)
- Salmon Pool (Named Stretch Of Inland Water, length: 120 meters)
- River Exe (Named Watercourse, length: 132,001 meters)

These tidal water bodies are located within and around the Exeter area. A map has been generated showing the Exeter city boundary and all main tidal water features. If you would like to view or download the map, let me know!

Agent: human

Where is tidal water near Exeter?

Agent: host_agent

Tool call: send_message - target: planning_agent - task_description: User wants to find tidal water near Exeter. Provide the general steps to solve this query.

Agent: planning_agent

The query is asking for tidal water near Exeter. The general area is Exeter, so we will focus on that. The entity to search for is tidal water, which may include rivers, estuaries, or coastal areas. We need to ensure we capture all relevant tidal water bodies in the vicinity of Exeter.

output steps: ["Find Exeter 1 area", "Search for tidal water bodies near Exeter as many entries as possible"]

Agent: host_agent

Tool call: send_message - target: named_area - task_description: Find the area polygon for Exeter.

Agent: named_area

Tool call: call_os_ngd - filters: None - bbox: None - polygon_or_point: True - street_address: None - filename: exeter_polygon

Tool call result: Search results have been found. Artifact generated is : exeter_polygon, Description is : A geopandas dataframe containing named area data with no filters applied for the query (so remember no search is performed here so spurious entities present) Find the area polygon for Exeter., count of records fetched : 98.

Agent: host_agent

Tool call: send_message - target: data_analysis_agent - task_description: Filter the artifact 'exeter_polygon' to find the specific polygon for Exeter. Save the filtered results as 'filtered_exeter_polygon'.

Agent: data_analysis_agent

Tool call: generate_metadata_for_artifacts - artifact_names: ['exeter_polygon']

Tool Code Executor

```
import matplotlib
matplotlib.use('Agg')
def filter_exeter_polygon(data: list):
    gdf = data[0]
    # Filter for polygons where name1_text is 'Exeter' and description is 'City', 'Town', 'Settlement', or 'Part Of Settlement'
    filtered = gdf[(gdf['name1_text'].str.lower() == 'exeter') &
                   (gdf['description'].isin(['City', 'Town', 'Settlement', 'Part Of Settlement']))]
    summary = f"Filtered to {len(filtered)} polygon(s) for Exeter with relevant settlement descriptions."
    artifact_name = 'filtered_exeter_polygon'
    artifact_description = 'GeoDataFrame containing only the polygon(s) for Exeter, filtered by name and settlement-related descriptions.'
    return [summary, artifact_name, artifact_description, filtered]
```

Tool Code Executor

```
import matplotlib
matplotlib.use('Agg')
def filter_exeter_polygon(data: list):
```

```
gdf = data[0]
# Filter for polygons where name1_text is 'Exeter' and description is 'City', 'Town', 'Settlement', or 'Part Of Settlement'
filtered = gdf[(gdf['name1_text'].str.lower() == 'exeter') &
               (gdf['description'].isin(['City', 'Town', 'Settlement', 'Part Of Settlement']))]
summary = f"Filtered to {len(filtered)} polygon(s) for Exeter with relevant settlement descriptions."
artifact_name = 'filtered_exeter_polygon'
artifact_description = 'GeoDataFrame containing only the polygon(s) for Exeter, filtered by name and settlement-related descriptions.'
return [summary, artifact_name, artifact_description, filtered]
```

Filtered to 1 polygon for Exeter with relevant settlement descriptions. An artifact has been generated with the name 'filtered_exeter_polygon'. This GeoDataFrame contains only the polygon(s) for Exeter, filtered by name and settlement-related descriptions.

Agent: named_area

The area polygon for Exeter has been successfully filtered and saved. The artifact name is filtered_exeter_polygon, which contains only the polygon(s) for Exeter, filtered by name and settlement-related descriptions.

Agent: host_agent

Tool call: send_message - target: water_features - task_description: Search for tidal water features near Exeter once the area polygon is found.

Agent: water_features

Tool call: call_os_ngd - filters: ['Sea', 'Tidal Ford'] - bbox: 'Exeter' - polygon_or_point: True - street_address: False - filename: 'exeter_tidal_water_features'

Tool call result: There was an error while calling NGD. Most probable cause is wrong artifact name for bbox parameter. Try a few times with the correct artifact names.

Tool call: call_os_ngd - filters: ['Sea', 'Tidal Ford'] - bbox: 'Exeter Area' - polygon_or_point: True - street_address: False - filename: 'exeter_area_tidal_water_features'

Tool call result: There was an error while calling NGD. Most probable cause is wrong artifact name for bbox parameter. Try a few times with the correct artifact names.

Tool call: call_os_ngd - filters: None - bbox: None - polygon_or_point: True - street_address: False - filename: 'available_artifacts'

Tool call result: Multiple search results have been found in multiple datasets. A summary of each is provided. Artifacts generated are : ['waterpoint_available_artifacts', 'water_available_artifacts'], Descriptions are : ['A geopandas dataframe containing water point data with bbox applied for the query {query} with filters applied {filters}', 'A geopandas dataframe containing water data with bbox applied for the query {query} with filters applied {filters}'], counts of records fetched are : [580, 313].

Tool call: call_os_ngd - filters: ['Sea', 'Tidal Ford'] - bbox: 'water_available_artifacts' - polygon_or_point: True - street_address: False - filename: 'exeter_tidal_water_features'

Tool call result: No results found for your search terms

I couldn't find any tidal water features near Exeter. If you have any other queries or need further assistance, feel free to ask!

Agent: host_agent

Tool call: send_message - target: water_network - task_description: Search for tidal water features (such as tidal rivers, estuaries, or tidal stretches) within the 'filtered_exeter_polygon' area.

Agent: water_network

Tool call: call_os_ngd - filters: ['tidal water features'] - bbox: 'filtered_exeter_polygon' - polygon_or_point: True - street_address: False - filename: 'tidal_water_features_exeter'

Tool call result: Multiple search results have been found in multiple datasets. A summary of each is provided. Artifacts generated are : ['waterlinkset_tidal_water_features_exeter', 'waterlink_tidal_water_features_exeter'], Descriptions are : ["A geopandas dataframe containing water link set data with bbox applied for the query Search for tidal water features (such as tidal rivers, estuaries, or tidal stretches) within the 'filtered_exeter_polygon' area. with no filters applied for the query (so remember no search is performed here so spurious entities present).", "A geopandas dataframe containing water link data with bbox applied for the query Search for tidal water features (such as tidal rivers, estuaries, or tidal stretches) within the 'filtered_exeter_polygon' area. with no filters applied for the query (so remember no search is performed here so spurious entities present)."], counts of records fetched are : [11, 430].

Agent: host_agent

Tool call: send_message - target: data_analysis_agent - task_description: Analyze the artifact 'waterlinkset_tidal_water_features_exeter' to filter out spurious entities and focus on tidal water features. Save the filtered results as 'filtered_tidal_water_features_exeter'.

Agent: data_analysis_agent

Tool call: generate_metadata_for_artifacts - artifact_names: ['waterlinkset_tidal_water_features_exeter']

Tool Code Executor

```
import matplotlib
matplotlib.use('Agg')
def filter_tidal_water_features_exeter(data: list):
    import geopandas as gpd
    # Extract the GeoDataFrame
    gdf = data[0]
    # Focus on tidal water features: likely candidates are 'Exeter Canal' and 'River Exe',
    # but also consider 'Named Canal', 'Named Watercourse', and 'Named Stretch Of Inland Water' in 'description'.
    # Remove spurious entities (e.g., small brooks, leats, or features not typically tidal)
    tidal_names = [
        'Exeter Canal',
        'River Exe',
        'Salmon Pool' # Salmon Pool is a known tidal pool in Exeter
    ]
    # Filter by both name and description for quality
    mask = (
        (gdf['name1_text'].isin(tidal_names)) &
        (gdf['description'].isin(['Named Canal', 'Named Watercourse', 'Named Stretch Of Inland Water']))
    )
    filtered = gdf[mask].copy()
    summary = f"Filtered tidal water features in Exeter: {len(filtered)} out of {len(gdf)} total entries. Names: {filtered['name1_text'].tolist()}"
    artifact_name = 'filtered_tidal_water_features_exeter'
    artifact_description = (
        "GeoDataFrame containing only the main tidal water features in Exeter, "
        "filtered by name and description to exclude minor or spurious entities. "
        "Columns preserved: all original columns."
    )
    return [summary, artifact_name, artifact_description, filtered]
```

Filtered tidal water features in Exeter: 3 out of 11 total entries. The filtered features are: Exeter Canal, Salmon Pool, and River Exe.

An artifact named 'filtered_tidal_water_features_exeter' has been generated. It contains a GeoDataFrame with only the main tidal water features in Exeter, filtered by name and description to exclude minor or spurious entities. All original columns are preserved.

Agent: host_agent

Tool call: send_message - target: data_analysis_agent - task_description: Analyze the artifact 'waterlink_tidal_water_features_exeter' to filter out spurious entities and focus on tidal water features. Save the filtered results as 'filtered_tidal_water_features_exeter_links'.

Agent: data_analysis_agent

Tool call: generate_metadata_for_artifacts - artifact_names: ['waterlink_tidal_water_features_exeter']

Tool Code Executor

```
import matplotlib
matplotlib.use('Agg')
def filter_tidal_water_features_exeter_links(data: list):
    import pandas as pd
    gdf = data[0]
    # Focus on tidal water features only, using 'watertype' and relevant descriptions
    tidal_mask = (
        (gdf['watertype'] == 'Tidal') &
        (gdf['theme'] == 'Water') &
        (gdf['description'].isin(['Watercourse', 'Canal', 'Still Water', 'Leat']))
    )
    # Remove spurious entities: require a name or secondary name, and primacy 1 (main features)
    named_mask = (
        (gdf['name1_text'].notnull()) | (gdf['namesecondary1_text'].notnull())
    ) & (gdf['primacy'] == '1')
    filtered = gdf[tidal_mask & named_mask].copy()
    summary = f"Filtered tidal water features links in Exeter: {len(filtered)} out of {len(gdf)} total entries. Only main, named tidal water features retained."
    artifact_name = 'filtered_tidal_water_features_exeter_links'
    artifact_description = (
        "GeoDataFrame containing only the main, named tidal water features links in Exeter. "
        "Filtered by 'watertype'=='Tidal', 'theme'=='Water', relevant 'description', and requiring a name and primacy 1. "
        "All original columns are preserved."
    )
    return [summary, artifact_name, artifact_description, filtered]
```

Filtered tidal water features links in Exeter: 2 out of 430 total entries. Only main, named tidal water features have been retained.

An artifact named 'filtered_tidal_water_features_exeter_links' has been generated. It contains a GeoDataFrame with only the main, named tidal water features links in Exeter, filtered by 'watertype'=='Tidal', 'theme'=='Water', relevant 'description', and requiring a name and primacy 1. All original columns are preserved.

Agent: water_network

The search and filtering process for tidal water features within the 'filtered_exeter_polygon' area has been completed. Here are the results:

- 1. **Filtered Tidal Water Features:**
- 2. **Artifact Name:** filtered_tidal_water_features_exeter
 - Description:** Contains the main tidal water features in Exeter, such as Exeter Canal, Salmon Pool, and River Exe. Spurious entities have been excluded.
 - Filtered Tidal Water Features Links:**
- 5. **Artifact Name:** filtered_tidal_water_features_exeter_links
- 6. **Description:** Contains the main, named tidal water features links in Exeter. Filtered by relevant criteria to ensure accuracy.

These artifacts provide a refined view of the tidal water features in the specified area.

Agent: host_agent

Tool call: send_message - target: plotting_agent - task_description: Plot the main tidal water features near Exeter using the artifacts 'filtered_exeter_polygon' and 'filtered_tidal_water_features_exeter'. Provide a summary of the main tidal water bodies found.

Agent: plotting_agent

Tool call: generate_metadata_for_artifacts - artifact_names: ['filtered_exeter_polygon', 'filtered_tidal_water_features_exeter']

Tool Code Executor

```

import matplotlib
matplotlib.use('Agg')
def plot_main_tidal_water_features_exeter(data: list):
    import geopandas as gpd
    import folium
    from shapely.geometry import mapping

    # Unpack data
    exeter_polygon = data[0]
    tidal_water_features = data[1]

    # Ensure CRS is EPSG:4326 for folium
    if exeter_polygon.crs != 'EPSG:4326':
        exeter_polygon = exeter_polygon.to_crs('EPSG:4326')
    if tidal_water_features.crs != 'EPSG:4326':
        tidal_water_features = tidal_water_features.to_crs('EPSG:4326')

    # Get centroid of Exeter for map centering
    centroid = exeter_polygon.geometry.iloc[0].centroid
    m = folium.Map(location=[centroid.y, centroid.x], zoom_start=13, tiles='cartodbpositron')

    # Plot Exeter boundary
    folium.GeoJson(
        exeter_polygon.geometry.iloc[0],
        name='Exeter Boundary',
        style_function=lambda x: {'color': 'black', 'weight': 2, 'fill': False}
    ).add_to(m)

    # Plot tidal water features
    color_dict = {'Exeter Canal': 'blue', 'Salmon Pool': 'purple', 'River Exe': 'cyan'}
    feature_summary = []
    for idx, row in tidal_water_features.iterrows():
        name = row['name1_text']
        desc = row['description']
        length = row['geometry_length']
        color = color_dict.get(name, 'blue')
        folium.GeoJson(
            row['geometry'],
            name=f"{name} ({desc})",
            style_function=lambda x, color=color: {'color': color, 'weight': 4}
        ).add_to(m)
        feature_summary.append(f"{name} ({desc}, length: {length:.1f}m)")

    # Add layer control
    folium.LayerControl().add_to(m)

    # Save map
    map_filename = 'main_tidal_water_features_exeter.html'

```



```

m.save(map_filename)

# Prepare summary
summary = (
    f"Main tidal water features near Exeter:\n" +
    "\n".join(feature_summary[:5]) +
    f"\nTotal features: {len(tidal_water_features)}. Map shows Exeter boundary and all main tidal water bodies."
)

return [
    summary,
    'main_tidal_water_features_exeter_map',
    'A folium map showing the main tidal water features (Exeter Canal, Salmon Pool, River Exe) near Exeter, overlaid with the city boundary.',
    map_filename
]

```

Main tidal water features near Exeter: Exeter Canal (Named Canal, length: 8281.0m) Salmon Pool (Named Stretch Of Inland Water, length: 120.1m) River Exe (Named Watercourse, length: 132001.1m) Total features: 3. Map shows Exeter boundary and all main tidal water bodies.

Artifact name: main_tidal_water_features_exeter_map

Artifact description: A folium map showing the main tidal water features (Exeter Canal, Salmon Pool, River Exe) near Exeter, overlaid with the city boundary.

Artifact data: main_tidal_water_features_exeter.html